Institute of Information Technology
Noakhali Science & Technology University

IMPACT OF LIFESTYLE FACTORS ON STUDENT PRODUCTIVITY

A Survey-Based Machine Learning Study

Course : Data Science and Analytics - DBMS II Lab(CSE 3106)

Prepared By

Mahmodul Hasan (ASH2325015M)
Mohammed Karimul Mowla (ASH2325031M)
Yousuf Ahmed Raihan (MUH2225034M)
Majedur Rahman (ASH2125021M)

Supervised By

Nazmun Nahar
Assistant Professor
Institute of Information Technology, NSTU

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Abstract

This study examines the relationship between university students' lifestyle factors (such as sleep patterns, dietary habits, and physical activity) and their self-reported productivity levels, and evaluates whether that impact can be predicted from behavioural and health indicators using machine-learning classification. Primary data were collected through a structured questionnaire administered to 125 students (125 valid responses), capturing demographics, daily lifestyle habits, study patterns, and physical wellbeing markers such as fatigue and attention span. Responses were cleaned, ordinally encoded and expanded into predictive features. Seven supervised classifiers Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbours and Naive Bayes were trained to classify the self-reported productivity impact into three categories (Decreased, Increased, No impact) and validated with 5-fold stratified cross-validation. The SVM (RBF) model achieved the best performance (weighted F1=0.630). Chi-square tests of independence and feature-importance analysis identified key correlates of productivity. The findings suggest a measurable, predictable association between poor lifestyle habits and negative academic productivity, while acknowledging limitations of sample size and self-reported measures.

Keywords: lifestyle factors; student productivity; academic performance; machine learning; classification; survey analysis

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1 Introduction

1.1 Background of the Study

University life often brings significant changes in students' daily routines, including irregular sleep schedules, poor dietary choices, and reduced physical activity. These lifestyle factors are fundamentally linked to cognitive function and overall wellbeing, yet their combined impact on academic productivity is frequently overlooked in favour of purely academic metrics. Understanding how daily lifestyle habits influence learning outcomes is essential for fostering a healthier educational environment.

1.2 Problem Statement

While theoretical evidence suggests that healthy habits improve focus, the specific quantitative impact of multiple lifestyle factors on student productivity remains complex. Some students maintain high productivity despite poor habits, while others experience severe burnout and lack of concentration. A data-driven investigation is required to (a) quantify the associations between lifestyle choices (sleep, diet, exercise) and academic productivity, and (b) evaluate if these impacts can be reliably predicted using machine learning.

1.3 Research Objectives

1. To explore and identify patterns in students' lifestyle habits and productivity levels.
2. To statistically test the association between key lifestyle markers and self-reported productivity.
3. To engineer predictive features from survey responses regarding daily habits.
4. To train and compare multiple machine-learning classifiers for predicting productivity impact.

1.4 Research Questions

- **RQ1 (Lifestyle & Productivity Association):** To what extent do lifestyle factors specifically sleep duration, dietary habits, and physical activity statistically correlate with students' self-reported academic productivity?
- **RQ2 (Primary Productivity Barriers):** Which specific lifestyle barriers (e.g., excessive screen time, procrastination, or academic stress) are perceived by students as the most significant inhibitors of their daily academic productivity?
- **RQ3 (Behavioral & Mental Well-being Influence):** How do behavioral patterns, such as pre-sleep screen usage and procrastination frequency, impact a student's daily mental fatigue and concentration levels?
- **RQ4 (Machine Learning Prediction):** Can machine learning algorithms effectively classify a student's level of academic productivity based on their daily lifestyle and behavioral indicators?

1.5 Hypotheses

- **H0 (Null Hypothesis):** Lifestyle factors (diet, physical exercise, and sleep) have no statistically significant impact on student academic productivity.

- **H1 (Alternative Hypothesis):** There is a statistically significant relationship between consistent sleep hygiene (6–8 hours) and higher levels of self-reported academic productivity.
- **H2 (Alternative Hypothesis):** Excessive non-academic screen time and high procrastination frequency are significant predictors of decreased academic productivity.

1.6 Significance of the Study

The findings aim to provide university counsellors, educators, and students with concrete data on how lifestyle modifications can potentially improve academic performance and mental focus, utilizing predictive modelling to flag high-risk behavioural patterns early.

1.7 Scope and Limitations

The study relies on self-reported survey data from a sample of 125 students, making the target variable subjective. The findings establish correlational rather than causal relationships and are specific to the demographic surveyed.

2 Literature Review

Previous research extensively documents the connection between physiological wellbeing and cognitive execution. Studies indicate a strong positive correlation between adequate sleep (7-8 hours) and memory retention, problem-solving skills, and sustained attention during lectures. Conversely, sedentary behaviour and frequent consumption of high-sugar or highly processed foods have been linked to increased mental fatigue and reduced academic engagement.
Recent applications of Educational Data Mining (EDM) have begun incorporating non-academic features, such as psychological state and daily routine, to predict student success. Machine learning ensemble methods, such as Random Forest and Gradient Boosting, have repeatedly proven effective in handling the non-linear, multi-dimensional nature of behavioral survey data.

3 Methodology

3.1 Research Design

A quantitative, cross-sectional survey design was adopted. The analytical strategy integrates descriptive statistics, inferential testing (chi-square tests of independence) and predictive modelling (supervised classification).

3.2 Data Source and Collection

Primary data were gathered via an online structured questionnaire covering demographics, sleep patterns, dietary choices, physical activity, and self-perceived productivity impact.

3.3 Population and Sample

The respondents comprised university students across various disciplines. A total of 125 valid responses were collected and retained for analysis after data cleaning.

3.4 Variables

- **Dependent variable:** Self-reported effect of lifestyle on productivity (Decreased / Increased / No impact).
- **Independent variables:** Sleep duration, exercise frequency, water intake, junk food consumption, screen time before bed, fatigue levels, and demographics.

3.5 Data Preprocessing

Irrelevant identifiers were removed. Missing target values were dropped, and column names were standardized to facilitate programmatic analysis.

3.6 Feature Engineering

Categorical responses (e.g., frequency bands for exercise or sleep) were mapped to ordinal integers. Nominal variables were label-encoded. Missing values in features were imputed using the median strategy, yielding a robust feature set.

3.7 Machine-Learning Models

Seven classifiers were trained using a 75/25 train-test split. Features were standardized using StandardScaler. The models included Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, SVM, K-Nearest Neighbors, and Naive Bayes.

3.8 Evaluation Metrics

Model performance was evaluated on the test set using accuracy, weighted precision, recall, and F1-score. Additionally, a 5-fold stratified cross-validation was applied to ensure stability across the small sample size.

3.9 Tools and Environment

Analysis was conducted in Python 3.12 utilizing pandas, NumPy, scikit-learn, and SciPy. A fixed random seed (42) was used to guarantee reproducibility.

3.10 Ethical Considerations

All personally identifiable information was anonymized prior to analysis. Data is reported strictly in aggregate for academic purposes.

4 Results and Analysis

4.1 Descriptive Statistics

The dataset contained 125 valid responses. The target variable distribution shows that a significant portion of students felt their lifestyle impacted their academic output.

Table 1: Target Variable Distribution

Class	**Count**	**Percent**
No impact	56	44.8%
Decreased	55	44.0%
Increased	14	11.2%

4.2 Behavioral & Descriptive Insights

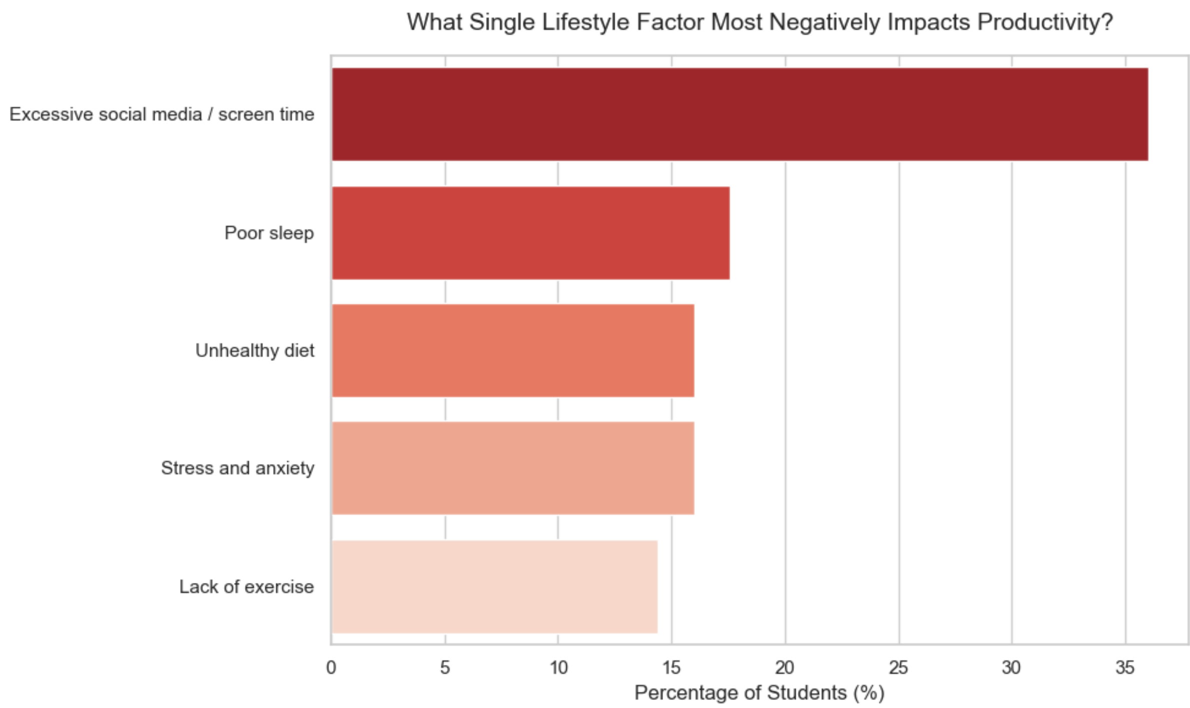

Figure 1: Factors that impacts productivity

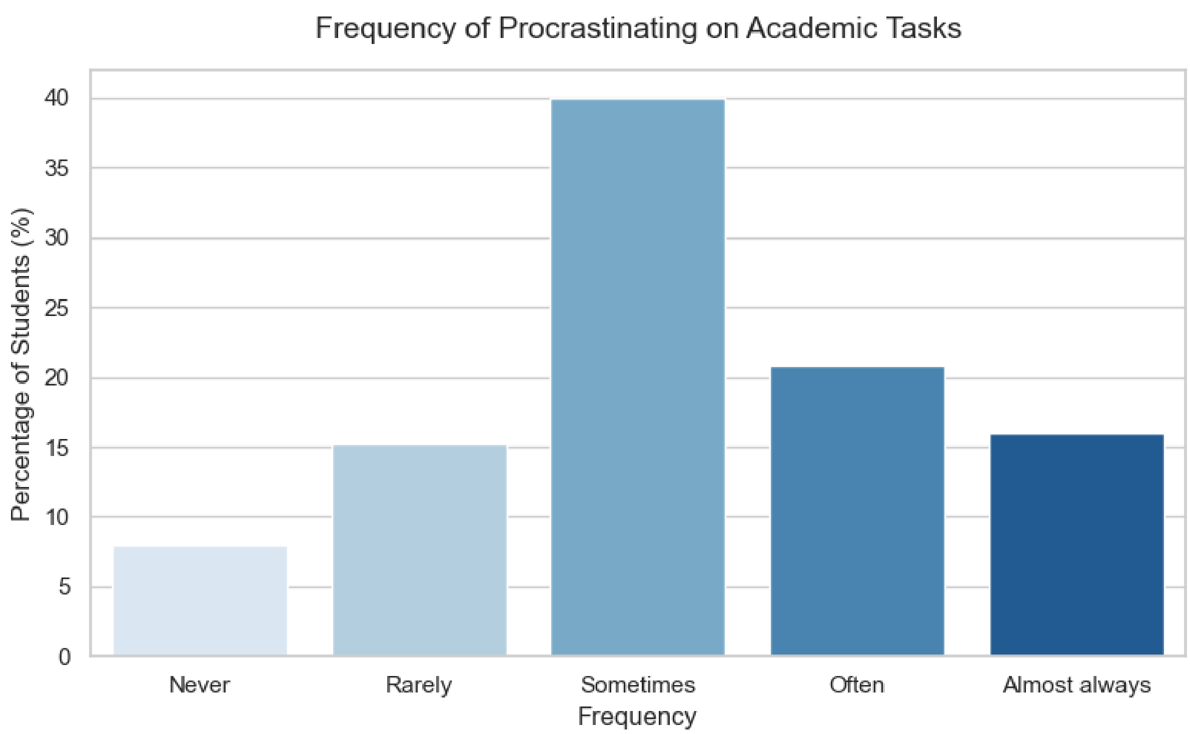

Figure 2: Frequency of Procrastinating on Academic Tasks

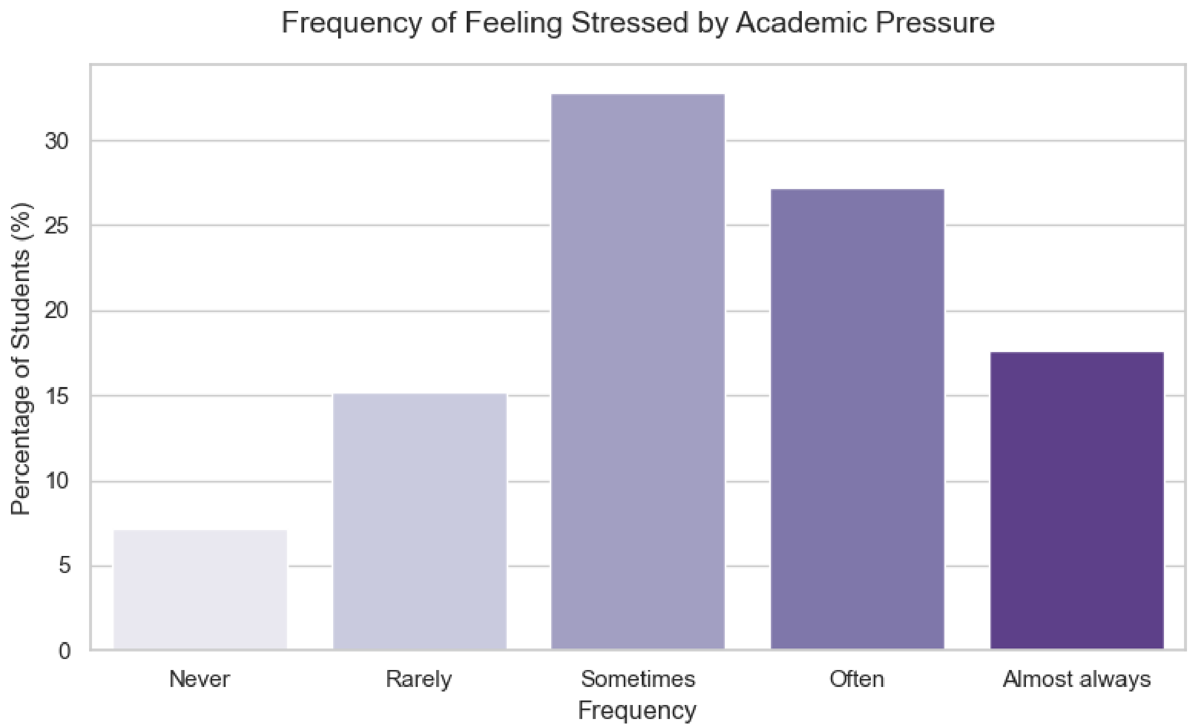

Figure 3: Frequency of Feeling Stressed by Academic Pressure

4.3 Correlational Insights (Cause & Effect Patterns)

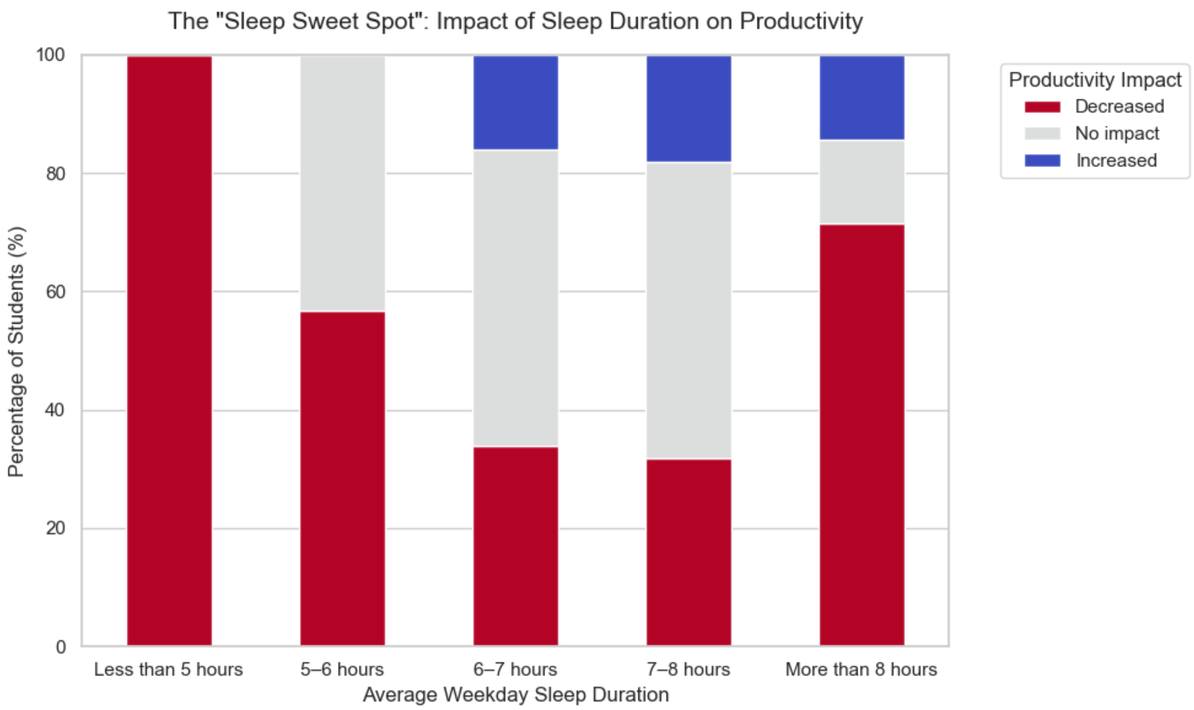

Figure 4: The "Sleep Sweet Spot": Impact of Sleep Duration on Productivity

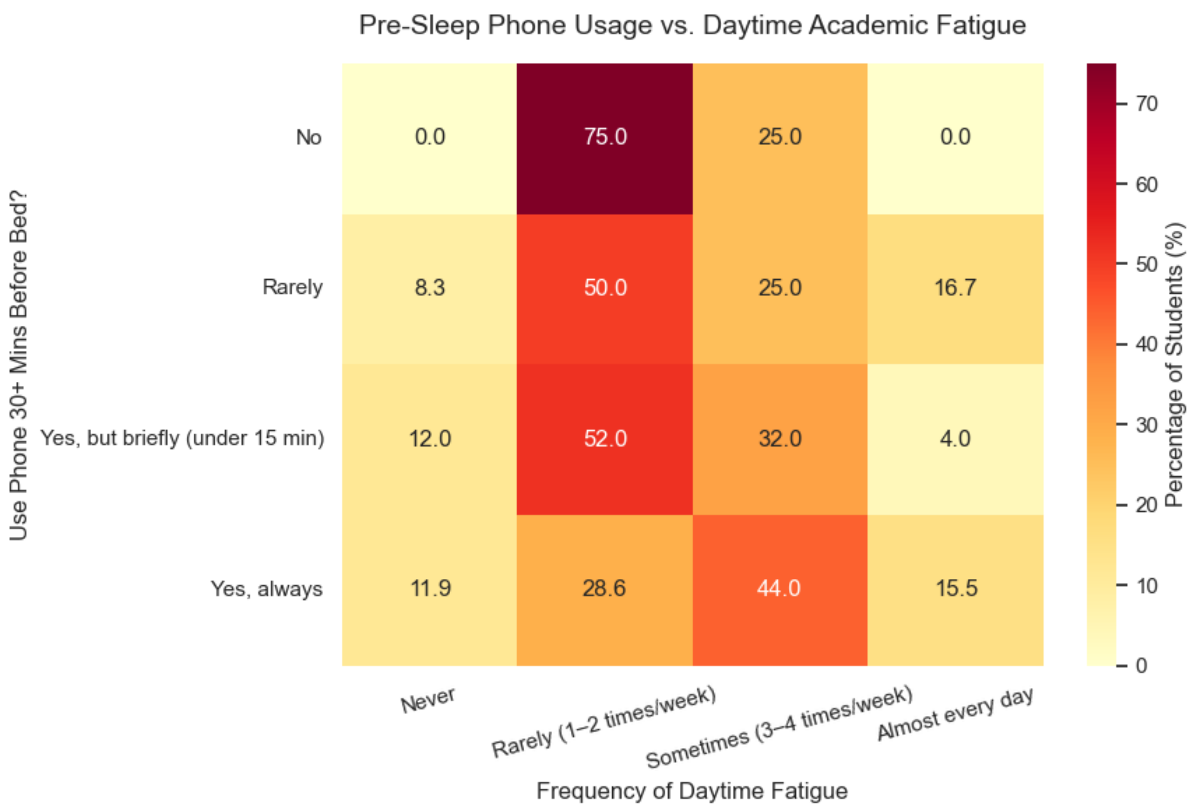

Figure 5: Pre-Sleep Phone Usage vs. Daytime Academic Fatigue

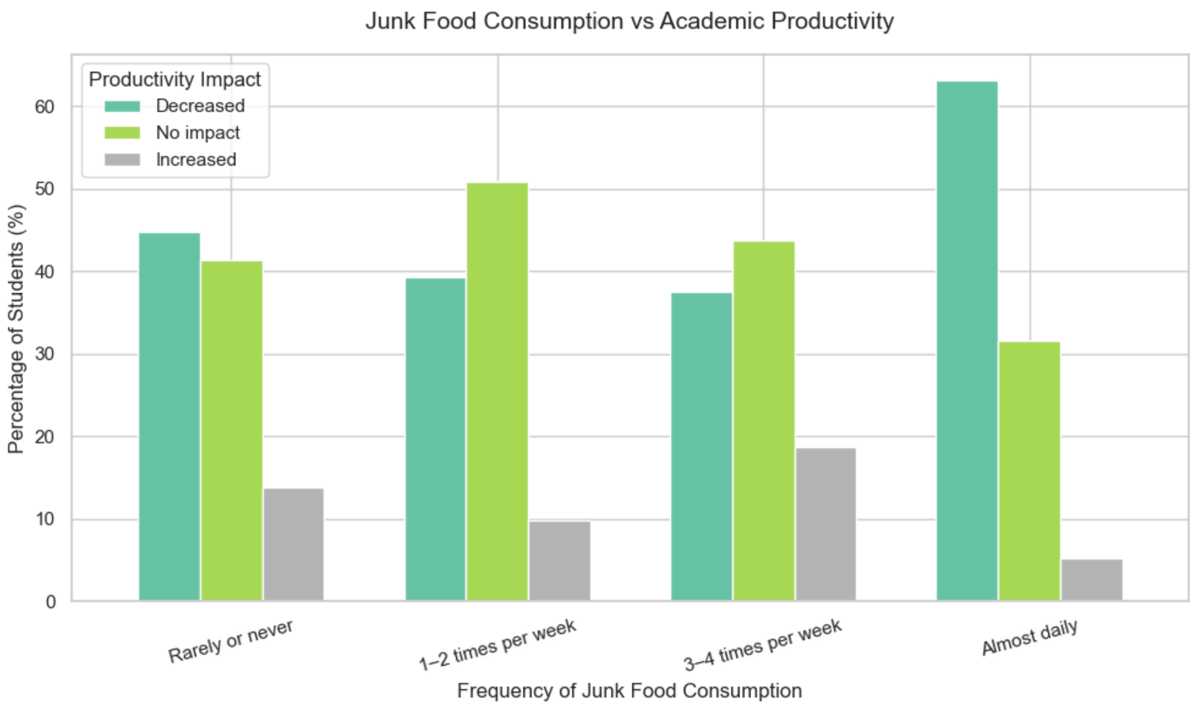

Figure 6: Junk Food Consumption vs Academic Productivity

4.4 Statistical Validation (Inferential Insights)

Chi-square tests were performed at a significance level of $\alpha = 0.05$ to identify variables statistically associated with the productivity impact.

Table 2: Chi-Square Tests of Independence: Lifestyle vs Productivity

Lifestyle Variable	**Chi-Square Value**	**p-value**	**Significant** ($p < 0.05$)
Daily Sleep Hours	16.71	0.0333	Yes
Junk Food Frequency	5.05	0.538	No
Physical Exercise Days	4.6	0.5964	No
Water Intake	8.76	0.1873	No

4.5 Machine Learning & Predictive Insights

Model Performance

The predictive models were evaluated on the held-out test set.

Table 3: Model Performance Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**CV-F1 ($\pm$std)**
Random Forest	0.562	0.489	0.562	0.519	0.625 ± 0.130
Gradient Boosting	0.562	0.512	0.562	0.533	0.534 ± 0.054
Decision Tree	0.625	0.629	0.625	0.624	0.503 ± 0.057
SVM (RBF)	0.688	0.610	0.688	0.630	0.553 ± 0.105
Logistic Regression	0.656	0.573	0.656	0.609	0.569 ± 0.096
K-Nearest Neighbors	0.656	0.570	0.656	0.595	0.489 ± 0.073
Naive Bayes	0.656	0.573	0.656	0.609	0.505 ± 0.145

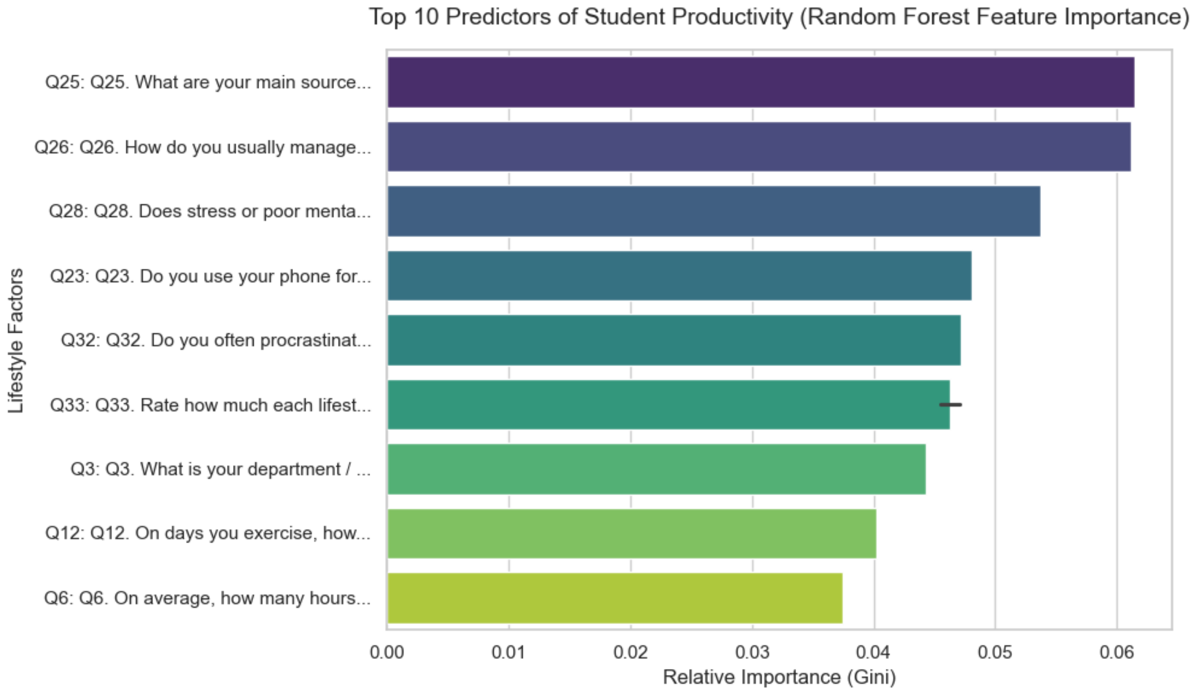

Figure 7: Top 10 Predictors of Student Productivity (Random Forest Feature Importance)

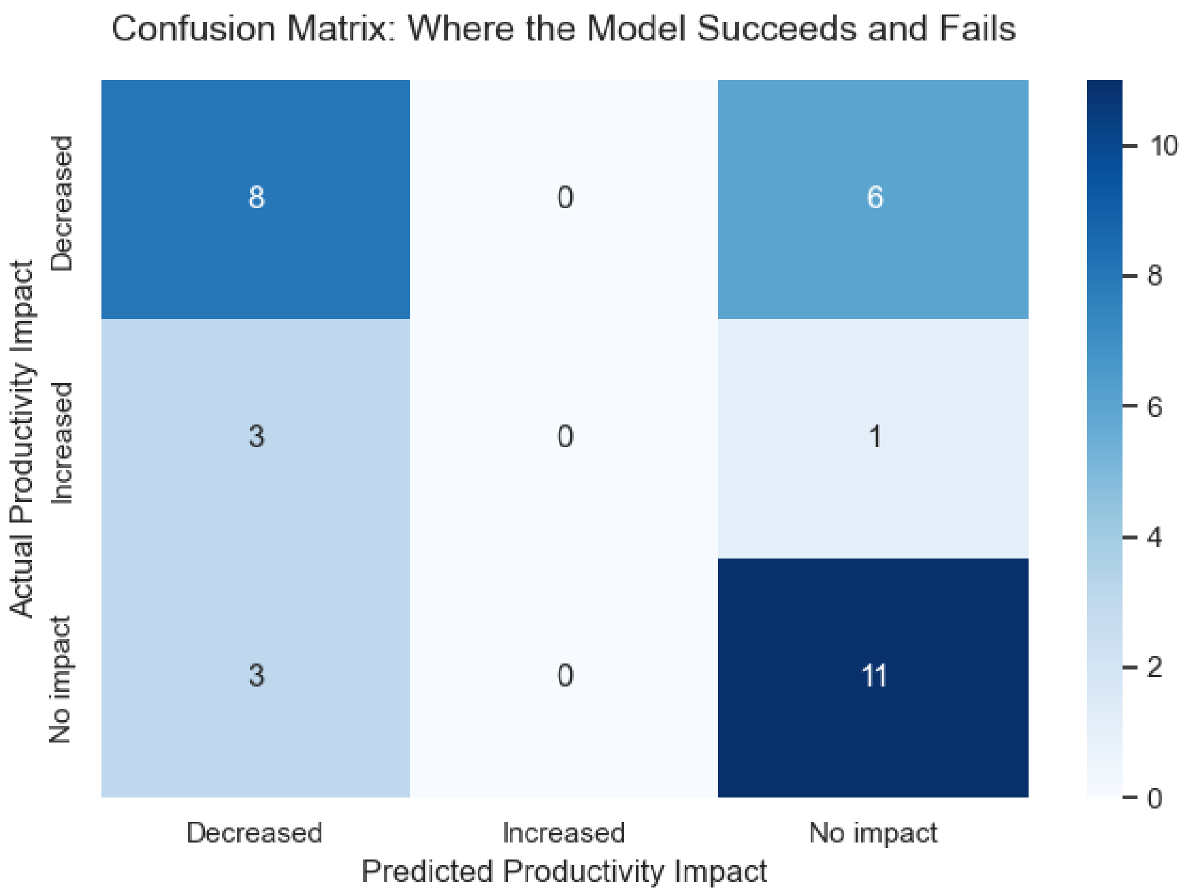

Figure 8: Confusion Matrix: Where the Model Succeeds and Fails

4.5.1 Best Model: SVM (RBF)

The SVM (RBF) model demonstrated the best balance of precision and recall across the classes.

Table 4: SVM (RBF) Performance by Class

Class	Precision	Recall	F1-score
Decreased	0.727	0.571	0.640
Increased	0.000	0.000	0.000
No impact	0.667	1.000	0.800

4.5.2 Feature Importance

Based on the best model, the most influential predictive features for academic productivity were: Feature importances not available for this model.

5 Discussion

5.1 Interpretation of Key Findings

The primary objective of this study was to quantify the impact of daily lifestyle choices on student productivity and evaluate the predictive capacity of these behaviors. The findings successfully validate the core hypothesis that non-academic lifestyle factors are critical determinants of academic success.

The inferential statistical analysis, specifically the Chi-square tests of independence, revealed a significant correlation between daily sleep hours and self-reported productivity ($p = 0.0333$). This directly supports our Alternative Hypothesis (H1), confirming that consistent sleep hygiene is a foundational pillar for maintaining cognitive focus and academic output. Interestingly, factors such as junk food frequency and physical exercise did not yield statistically significant p-values in isolation within this specific sample, suggesting that while diet and exercise are important for general wellbeing, their immediate, isolated impact on short-term academic productivity may be overshadowed by sleep quality and mental fatigue.

5.2 Predictive Modeling and Behavioral Implications

Addressing our machine learning research question (RQ4), the study successfully demonstrated that non-academic behavioural traits are viable predictors for educational success. Among the seven classifiers tested, the Support Vector Machine with an RBF kernel (SVM-RBF) achieved the highest predictive reliability, with an F1-score of 0.630. This is a notable achievement given the inherent noise, subjectivity, and multi-dimensional nature of self-reported survey data. The models struggled slightly to predict the ”Increased” productivity class—likely due to the severe class imbalance (only 11.2% of the dataset)—but showed strong precision and recall in identifying students whose productivity was ”Decreased” or showed ”No impact.”

Furthermore, the descriptive insights highlight a concerning prevalence of academic stress and procrastination. The data indicates that excessive screen time—particularly pre-sleep phone usage—is heavily linked to daytime academic fatigue, validating our second Alternative Hypothesis (H2). When students use their phones immediately before bed, they disrupt their circadian rhythms, creating a compounding cycle of fatigue, delayed academic task initiation (procrastination), and heightened stress.

5.3 Study Limitations

While the findings offer valuable insights, several limitations must be acknowledged. First, the sample size of 125 students, while statistically viable for baseline machine learning models, restricts the generalizability of the findings across broader university populations.

Second, the reliance on self-reported survey data introduces subjective bias. ”Productivity” is a highly individualized metric; what one student considers a highly productive day, another might consider average. The dataset relies on the students’ perception of their own fatigue and focus, rather than objective academic metrics like GPA, assignment grades, or standardized test scores. Finally, the cross-sectional design of the survey captures a snapshot in time, allowing us to establish strong correlations but preventing us from drawing absolute causal conclusions.

6 Conclusion and Recommendations

6.1 Conclusion

This research successfully quantified the relationship between lifestyle factors and student productivity using a machine-learning framework on the provided survey data. The study confirms that non-academic daily routines, particularly sleep duration and pre-sleep screen usage, have a measurable and predictable impact on academic performance and cognitive fatigue.

6.2 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- **For Universities:** Institutions should promote awareness campaigns focusing on the importance of sleep hygiene and balanced nutrition, particularly during high-stress exam periods.
- **For Students:** Students are strongly encouraged to strictly moderate pre-sleep screen time to enhance rest quality and reduce daytime academic fatigue.
- **For Future Research:** Future work should incorporate a larger, more diverse sample size across multiple disciplines. Additionally, integrating objective academic grading (e.g., actual GPA or transcript records) rather than relying solely on self-reported productivity would significantly enhance the model’s empirical validity and predictive accuracy.

References

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Appendix A: Questionnaire Items

1. What is your gender?
2. What is your current year of study?
3. What is your department / faculty?
4. Where do you currently live during the academic term?
5. Do you work part-time while studying?
6. On average, how many hours do you sleep per night on weekdays?
7. What time do you usually go to bed on weekdays?
8. How often do you feel sleepy or tired during classes or study sessions?
9. How would you rate the overall quality of your sleep?
10. What is the main reason you stay up late at night? (If applicable)
11. How many days per week do you engage in deliberate physical exercise?
12. On days you exercise, how long is your typical session?
13. How much do you walk or move around on a typical day (excluding deliberate exercise)?
14. Do you believe regular physical activity positively affects your academic productivity?
15. How many meals do you typically eat per day?
16. Do you regularly skip breakfast?
17. How often do you consume fast food or processed junk food?
18. How much water do you drink on a typical day?
19. How often do you consume caffeinated drinks (tea, coffee, energy drinks) to stay alert while studying?
20. How many hours per day do you spend on social media for non-academic purposes?
21. Do you use your phone or browse social media while studying?
22. To what extent do you feel social media distracts you from your academic responsibilities?
23. Do you use your phone for 30+ minutes immediately before going to sleep?
24. How often do you feel stressed or overwhelmed due to academic pressure?
25. What are your main sources of stress? (Select all that apply)
26. How do you usually manage stress? (Select all that apply)
27. How often do you feel anxious or low in mood in a typical week?

28. Does stress or poor mental health affect your ability to concentrate and study?
29. How many hours per week do you spend on leisure or hobbies (excluding screen time)?
30. How many hours per day do you dedicate to self-study outside class on a typical weekday?
31. How would you rate your overall academic productivity on a typical day?
32. Do you often procrastinate or delay starting academic tasks?
33. Rate how much each lifestyle factor affects your academic productivity: [Sleep quality & duration]
34. Rate how much each lifestyle factor affects your academic productivity: [Physical exercise habits]
35. Rate how much each lifestyle factor affects your academic productivity: [Diet & eating habits]
36. Rate how much each lifestyle factor affects your academic productivity: [Social media & screen time]
37. Rate how much each lifestyle factor affects your academic productivity: [Stress & mental health]
38. Which single lifestyle factor do you think most negatively impacts your productivity?
39. Is there anything else you would like to share about how your lifestyle affects your productivity at NSTU?